Part C — Sequence models & attention

ml-ladder writeup · part C

2026-09-24

Contents

C1 — Char-level RNN and LSTM **2**
- The idea . . . 2
- The math . . . 3
- What we measure, and why . . . 3
- Results . . . 4
- What surprised me . . . 4
- Open threads . . . 4
- Further reading . . . 6

C2 — Seq2seq with and without attention **6**
- The idea . . . 6
- The math . . . 7
- What we measure, and why . . . 7
- Results . . . 7
- What surprised me . . . 8
- Open threads . . . 8
- Further reading . . . 8

C3 — A transformer from scratch **10**
- The idea . . . 10
- The math . . . 10
- What we measure, and why . . . 11
- Results . . . 11
- What surprised me — and the result I cannot explain away . . . 12
- Open threads . . . 12
- Further reading . . . 14

C4 — Fine-tuning a small ViT **14**
- The idea . . . 14
- The math . . . 14
- What we measure, and why . . . 15
- Results . . . 15
- What surprised me . . . 16
- Open threads . . . 16
- Further reading . . . 16

C5 — What changed since C3, and what it cost **18**
- The idea . . . 18

The math . . . 18
What we measure, and why . . . 19
What I built . . . 19
Results . . . 19
What surprised me . . . 20
Open threads . . . 22
Further reading . . . 22

Part C at scale: more context, more data, a bigger transformer **22**
The idea . . . 22
What we measure, and why . . . 23
Results . . . 23
What surprised me . . . 25
Open threads . . . 26

The decoders, actually run **26**
Two generations, one architecture . . . 27
Through the candle fork . . . 27

What part C settled **29**

Status: complete (C1–C5) · **Constraint:** subsets and short budgets; C1/C3 share every setting so they can be compared
Numbers: seeds 0–2, reference seed 0 · raw values in `data/c-metrics.json`
Reproduce: `ladder run C1 ... C4` (needs `uv sync --extra torch`), or `uv run python tools/writeup_figures.py C`
Hardware: this part's sweep ran on a Colab T4 — C4 is ~25 min/seed on four CPU cores against ~35 s on the GPU

Part B's models all see their whole input at once. Part C's have to decide what to *carry*: a recurrent cell squeezes the past into a fixed state, attention keeps everything and learns where to look, and a ViT applies the second answer to images. Four rungs, one question —

where does the information have to travel, and what does the architecture charge for the trip? A vanilla RNN charges a multiplicative factor per step, which is why its gradient dies. An LSTM buys an additive path, the same trick as B4's residual connection, one year earlier and in time rather than depth. Attention refuses to compress at all and pays quadratic cost instead. Every result below is a price for distance.

Notation: T is sequence length, h_t the hidden state at step t, $Q, K, V \in \mathbb{R}^{T \times d}$ the attention projections, and d_k the per-head width.

C1 — Char-level RNN and LSTM

Question. The LSTM is supposed to remember further than a vanilla RNN. How much further, and does the extra memory show up as a better language model?

The idea

A **language model** predicts the next symbol from the ones before it. Trained on characters of Shakespeare, it learns spelling, then words, then some grammar, all from the single task "guess the next character".

A **recurrent neural network** (**RNN**) does this by reading one character at a time and keeping a **hidden state** h_t, a vector that is supposed to summarise everything read so far. Each step mixes the new character into the state, $h_t = \tanh(W_x x_t + W_h h_{t-1} + b)$, and predicts the next character from h_t. The same weights are reused at every step, so the model handles any length of text.

The weakness is in that recurrence. Information from 50 characters ago has been pushed through W_h and tanh 50 times, and so has the gradient on its way back, so long-range signals fade. The **LSTM** (long short-term memory) adds a second state, the **cell** c_t, which is updated by *addition*: **gates** (small learned valves with outputs between 0 and 1) decide how much of the old cell to keep (forget gate), how much new content to write (input gate), and how much to expose as output (output gate).

Left: in a vanilla RNN, the only path from an early character to a late prediction runs through the same matrix at every step. Right: the LSTM's cell state (gold line) runs straight through each step and is only multiplied by the forget gate and added to. When the forget gate is near 1, information and gradient both pass through almost unchanged.

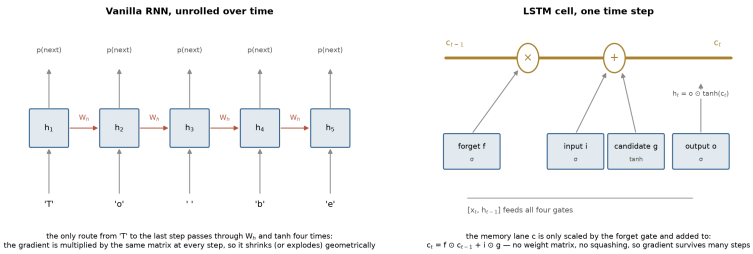


Figure 1: Unrolled RNN and one LSTM step

The math

Why the vanilla gradient dies. With $h_t = \tanh(W_x x_t + W_h h_{t-1} + b)$, backpropagation through time multiplies one Jacobian per step:

$$\frac{\partial h_T}{\partial h_t} = \prod_{i=t+1}^{T} \frac{\partial h_i}{\partial h_{i-1}} = \prod_{i=t+1}^{T} \operatorname{diag}(1 - h_i^2)\, W_h^\top .$$

Taking norms, $\|\partial h_T/\partial h_t\| \le (\gamma \|W_h\|)^{T-t}$ where $\gamma = \max_i \|\operatorname{diag}(1 - h_i^2)\| \le 1$ for tanh. The bound is **geometric in the distance**: if $\gamma\|W_h\| < 1$ the gradient vanishes exponentially, and if it exceeds 1 the gradient explodes instead. There is no setting of W_h that keeps a long-range signal alive without risking the other failure — which is the whole problem, and why the rungs clip gradients.

What the LSTM changes. The cell state's recurrence is additive:

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t \quad \Longrightarrow \quad \frac{\partial c_t}{\partial c_{t-1}} = \operatorname{diag}(f_t).$$

The product over steps becomes $\prod_i \operatorname{diag}(f_i)$ — no weight matrix, no saturating derivative, just the forget gates. A gate near 1 passes gradient through essentially unchanged, and the network *learns* which of them to open. That is why the forget-gate bias is initialized to 1: start by remembering.

Measuring it. The rung differentiates a scalar that depends only on the final state and reports $\|\partial L_T/\partial x_t\|$ against distance, then defines the **effective horizon** as the largest distance at which that norm is still above 10^{-6} of its value at the final position.

What we measure, and why

- **Bits per character** on held-out text — the average number of bits the model would need to encode each next character, i.e. cross-entropy in base 2. Lower is better. A model that knew nothing would need $\log_2 65 \approx 6$ bits (65 distinct characters). Good small models get near 2. It is the standard metric because it rewards being confidently right and punishes being confidently wrong.
- **Gradient reach**: the size of $\partial L/\partial x_t$ as a function of how far back x_t is, and the **effective horizon**, the furthest distance at which the gradient is still above a millionth of its final value. This measures the thing the LSTM was designed to fix directly, whatever the loss says.
- **Recall-probe accuracy** — a synthetic task (predict the *first* symbol after reading n filler

symbols) where only long-range memory helps. It isolates memory from everything else a language model uses. Chance is 1/8.

Results

Setup: tiny Shakespeare, 250k characters, 64-character context, 500 steps; medians over seeds 0–2.

model	parameters	bits/char	horizon	gradient at 63 steps back
vanilla RNN	98k	2.366	27	1.4e-11
vanilla RNN (width-matched)	343k	**2.263**	27	6.1e-11
LSTM	320k	2.334	**63**	9.4e-03

The horizon gap is the cleanest measurement in the part: 63 against 27 on every seed, and about 10^9 **times more gradient** at 63 steps back.

What surprised me

The LSTM loses the language-modelling comparison. At a matched parameter budget the width-matched vanilla RNN is 0.07 bits/char better, consistently across seeds (2.261–2.274 against 2.319–2.336, non-overlapping). The memory is real and does not pay, because next-character prediction in Shakespeare at a 64-character context is dominated by short-range structure — spelling, morphology, the line break after a speaker's name. Nothing in that needs to reach 63 steps back, and the LSTM spends four gates' worth of parameters per unit of state to buy reach the task does not use.

So the rung grew a second experiment: a **recall probe** where reach is the only thing that matters. The label is the sequence's first symbol, every later position is filler, and chance is 1/8:

sequence length	vanilla RNN	LSTM
5	1.000	1.000
15	0.127 (chance)	**1.000**
30	0.127	0.115 (chance)
60	0.127	0.115

There it is, and also its limit: the LSTM solves 15-step recall where the RNN has already collapsed to chance — about **3× the reach** — but at 30 steps it fails too, at this budget (120 steps, hidden 64). The horizon measurement says gradient *arrives* at 63 steps; the recall probe says arriving is not the same as learning to use it. I would rather report that than quietly pick the length where the LSTM still wins.

Open threads

GRUs and longer context have since been run on a GPU, in Part C at scale: on enwik8 the GRU is the best of the three recurrences at every context, and a 4.5M-parameter LSTM fails a copy task at 128 symbols. Still open: truncated BPTT over a context longer than one window, which is how these models were actually trained, and the recall probe with a proper budget sweep, to find where the LSTM's 63-step gradient becomes 63-step learning.

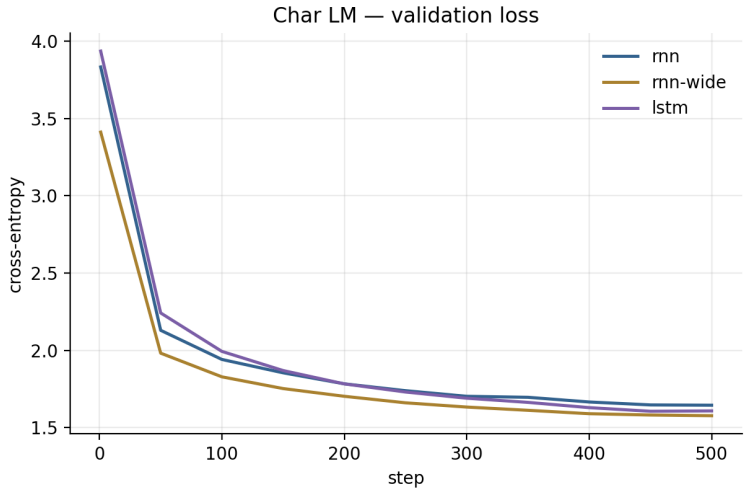


Figure 2: Validation loss

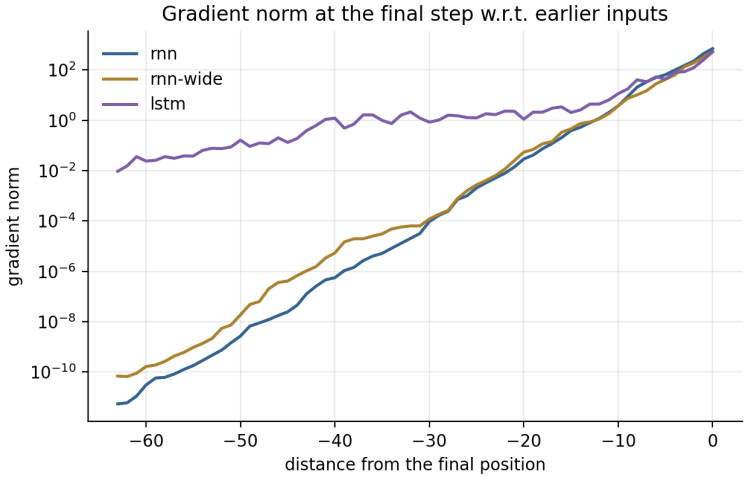


Figure 3: Gradient reach

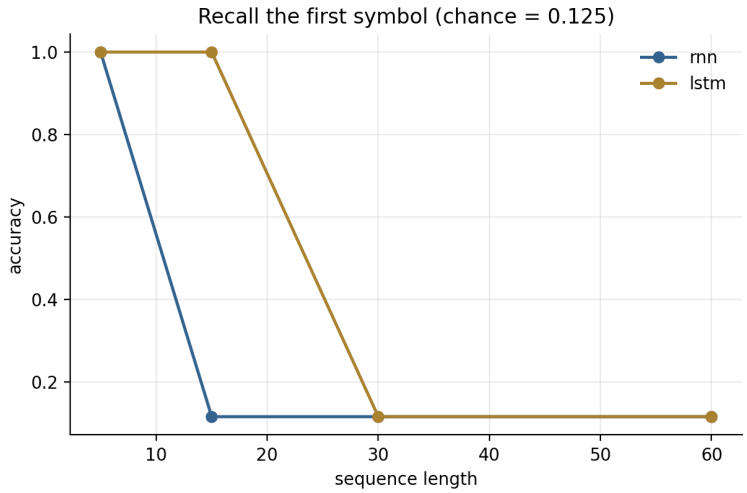


Figure 4: Recall accuracy

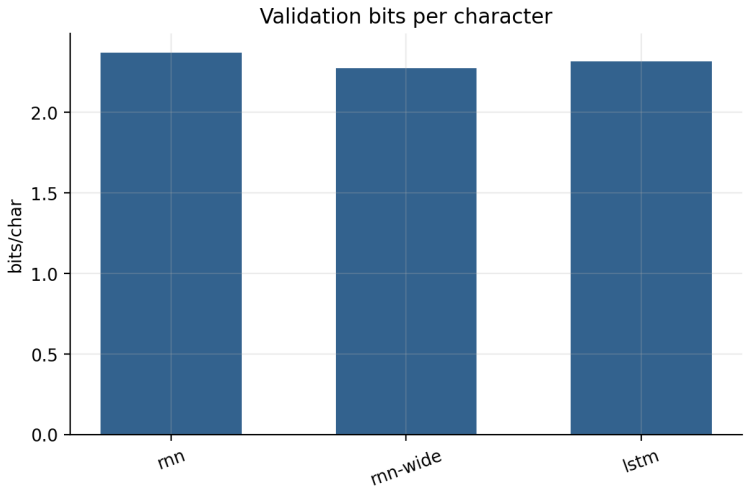


Figure 5: Bits per character

Further reading

Long Short-Term Memory (Hochreiter & Schmidhuber); Pascanu et al., *On the difficulty of training recurrent neural networks*; Karpathy, *The Unreasonable Effectiveness of RNNs*.

C2 — Seq2seq with and without attention

Question. How much does forcing a whole source sentence through one fixed-size vector actually cost?

The idea

Sequence-to-sequence (**seq2seq**) models map one sequence to another, such as a sentence to its translation, or here a date written any which way ("on the 3rd of May, 1998") to its ISO form ("1998-05-03"). An **encoder** RNN reads the input, and a **decoder** RNN writes the output one character at a time.

In the original design the encoder's *last* hidden state is the only thing handed to the decoder: the entire input must be compressed into one fixed-size vector, however long it is. **Attention** removes that bottleneck. At every output step the decoder scores each encoder state for relevance, turns the scores into weights that sum to 1 (softmax), and reads a weighted average. It can look straight at "May" when writing the month, instead of hoping the month survived compression.

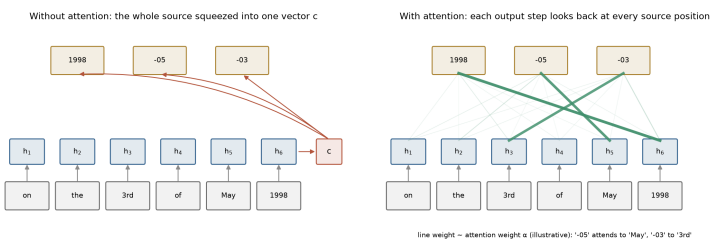


Figure 6: Seq2seq with and without attention

Left: every output step depends on the one vector c. Right: each output step draws on all source positions, weighted by learned relevance. The weights in this sketch are illustrative. The real ones, learned by the

rung, are plotted under Results.

The math

The bottleneck. A classic encoder-decoder compresses the source to a single vector $c = h_{T_s}$ and generates from it:

$$p(y_1, \dots, y_{T_t} \mid x) = \prod_j p\big(y_j \mid y_{<j},\, c\big)\,.$$

Everything the decoder can ever know about the source must fit in c, whose size is fixed while T_s grows. With d-dimensional states the channel carries $O(d)$ numbers regardless of input length — so the cost of the bottleneck should grow with the amount of *information* in the source, not its padded length. (The first version of this rung missed exactly that, swept padding, and found nothing.)

Attention removes the constraint. Keep every encoder state and build a fresh context per output step:

$$e_{ji} = \text{score}(s_j, h_i), \qquad \alpha_{ji} = \frac{\exp e_{ji}}{\sum_{i'} \exp e_{ji'}}, \qquad c_j = \sum_i \alpha_{ji} h_i.$$

This rung uses Luong's dot product, $\text{score}(s, h) = s^\top h$ (Bahdanau's additive $v^\top \tanh(W[s; h])$ is the other standard choice). The context is now a *convex combination of all source states*, so no fixed budget limits what reaches the decoder; the price is $O(T_s T_t)$ scores instead of $O(T_t)$, and it is the same price the transformer pays in C3.

Padding must be masked **before** the softmax, by setting those logits to $-\infty$: masking afterwards would leave the remaining weights un-normalized.

Why this task. Every output character of an ISO date is a copy of some input span, so the correct alignment is known in advance and the attention map can be *scored* — the fraction of output characters whose attention peak sits on a matching input character — rather than admired.

What we measure, and why

- **Exact-match accuracy** — the fraction of dates converted with every character right. This is the metric that matters for the task, since a date with one wrong digit is simply wrong.
- **Character-level accuracy** — fraction of individual characters right. It is reported *alongside* exact match because the gap between the two is informative: it shows a model producing well-formed but wrong answers.
- **Both, against source length** — the bottleneck argument predicts the fixed vector fails as the input grows, so length is the variable to sweep.
- **Alignment score** — for the attention model, the fraction of output characters whose strongest attention lands on the input characters they were copied from. The date task has a known correct alignment, so the attention map can be *graded* instead of just admired.

Results

Setup: date normalization, filler words around the date, medians over seeds 0–2.

filler	source length	bottleneck exact	attention exact	bottleneck char-level
none	18	0.997	1.000	0.999
10 words	101	0.326	1.000	0.899
24 words	193	**0.005**	**1.000**	0.710

The attention model is perfect at every length; the bottleneck model falls off a cliff. Alignment score 0.729 median — high, and *not* 1.0 by construction, since the month digits must come from a month *name* and count as misses.

What surprised me

The gap between the two accuracy measures. At 193 characters the bottleneck model still gets **71% of characters** right while getting **0.5% of dates** right. It has learned the *shape* of the answer — four digits, dash, two digits, dash, two digits, roughly the right ranges — and lost the content. That is the most honest picture of what a too-small channel does: not noise, but confident, well-formed, wrong output. Character accuracy would have made this look like a mild degradation; exact match shows it is total.

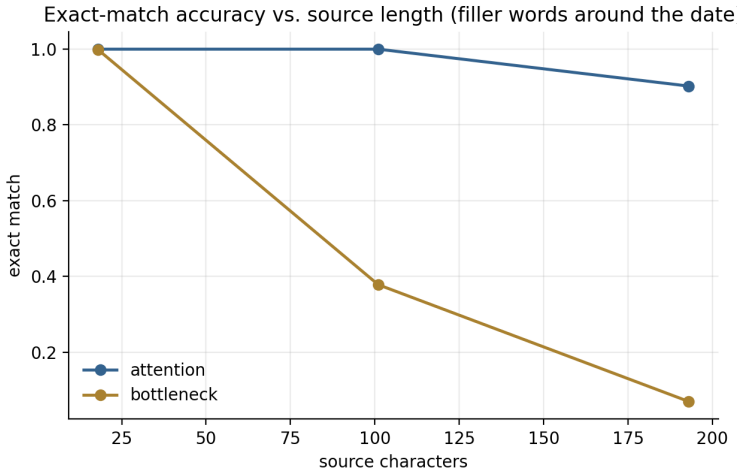


Figure 7: Accuracy by source length

Open threads

Bahdanau's additive score, which has an extra parameter matrix and is claimed to matter for longer sources. Coverage penalties, the standard fix for attention that revisits one position and drops another. And beam search instead of greedy decoding, which should matter most exactly where the bottleneck model is uncertain.

Further reading

Neural Machine Translation by Jointly Learning to Align and Translate (Bahdanau et al.); Effective Approaches to Attention-based NMT (Luong et al.); Sequence to Sequence Learning (Sutskever et al.).

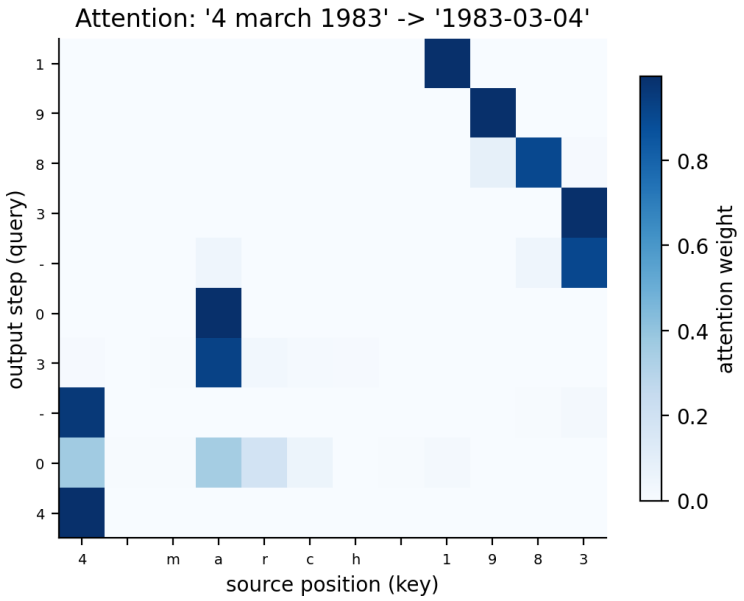


Figure 8: Attention map

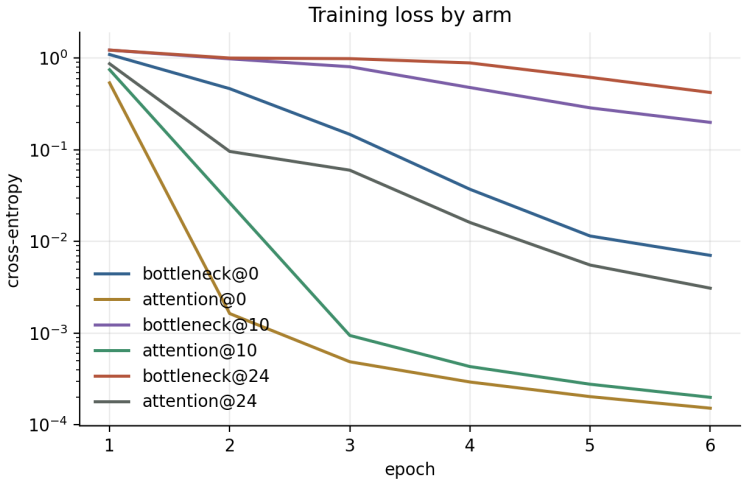


Figure 9: Training loss

C3 — A transformer from scratch

Question. Does the implementation actually respect causality — and how much of a transformer's ability is the positional encoding?

The idea

The **transformer** drops recurrence entirely. Every position looks at every other position directly through **self-attention**, so there is no chain of steps for information to fade along.

Each token's vector is projected three ways: a **query** ("what am I looking for?"), a **key** ("what do I contain?") and a **value** ("what do I pass on if chosen?"). A token's new representation is the average of all values, weighted by how well its query matches each key. **Multi-head** attention runs several of these in parallel with different projections, so one head can track syntax while another tracks, say, the previous character. Between attention layers, a per-token **MLP** processes what was gathered. **Residual connections** (B4) and **LayerNorm** keep the stack trainable.

Two details make it a *language model*:

- the **causal mask** stops a position from attending to later positions, so the model cannot cheat by reading the answer;
- a **positional encoding** is added to each token, because attention by itself is order-blind: it treats its input as a set, not a sequence.

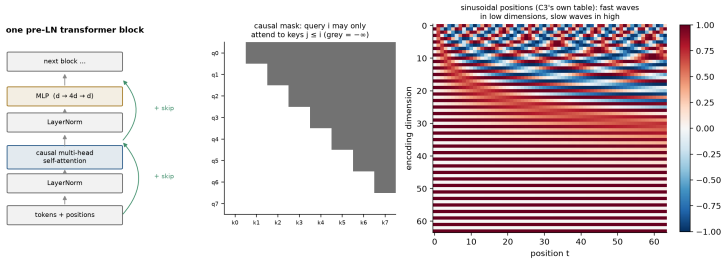


Figure 10: One transformer block, its causal mask and its positional encoding

Left: one block, as C3 builds it. Middle: the causal mask, where each query row may only see keys at or before it. Right: C3's actual sinusoidal encoding. Each dimension oscillates at its own frequency, so every position gets a unique pattern, and "k steps back" is the same rotation wherever you are.

The math

Scaled dot-product attention, and why the scale is there:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V.$$

If the entries of q and k are independent with zero mean and unit variance, then $q^\top k = \sum_{i=1}^{d_k} q_i k_i$ has variance d_k. Without the $1/\sqrt{d_k}$, logits grow like $\sqrt{d_k}$, the softmax saturates toward a one-hot distribution, and its Jacobian $\text{diag}(p) - pp^\top$ goes to zero — attention stops learning precisely when the model is made wider. Dividing by $\sqrt{d_k}$ holds the logit variance at 1 regardless of head width.

The causal mask is $M_{ij} = -\infty$ for $j > i$ and 0 otherwise, applied *before* the softmax so the forbidden positions get exactly zero weight rather than a small one. This is trivially easy to get

subtly wrong — off by one, or masked after the softmax — and a model that leaks a single future position still trains beautifully while being worthless. The rung therefore checks it: perturb the last token, and no earlier logit may move.

Positional encoding, and what happens without it. Self-attention is *permutation-equivariant*: for a permutation matrix P,

$$\text{Attention}(PQ, PK, PV) = P\,\text{Attention}(Q, K, V),$$

so a transformer with no positional information cannot distinguish "abc" from "cba" — with a causal mask it degenerates to a function of the *multiset* of preceding characters. The sinusoidal encoding

$$PE_{(t,2i)} = \sin\!\left(\frac{t}{10000^{2i/d}}\right), \qquad PE_{(t,2i+1)} = \cos\!\left(\frac{t}{10000^{2i/d}}\right)$$

fixes this with a useful extra property: for a fixed offset k, the pair $(\sin\omega t, \cos\omega t)$ maps to $(\sin\omega(t+k), \cos\omega(t+k))$ by a rotation that depends on k but **not** on t, so "k steps back" is a linear operation the attention can express uniformly across the sequence.

Pre-LN blocks. The rung normalizes inside the residual branch, $x \mapsto x + f(\text{LN}(x))$, rather than the 2017 paper's $x \mapsto \text{LN}(x + f(x))$. Post-LN puts a normalization on the skip path itself and needs learning-rate warmup to train at all; pre-LN keeps the identity path clean — B4's argument again.

What we measure, and why

- **Bits per character**, on the same text, split, windows and budget as C1, so the RNN, LSTM and transformer numbers are directly comparable.
- **The positional ablation** — the same model with learned, sinusoidal and *no* positional encoding. The theory says removing it should hurt badly, and measuring by how much turns the equivariance argument into a number.
- **The causal check** — perturb the last input token and confirm no earlier output moves, plus the attention weight above the diagonal. Both must be exactly 0. A leak here would make every other number meaningless, and a leaky model still trains beautifully, which is why it is tested rather than trusted.
- **Parameter count** for every arm, because "transformer vs LSTM" is only a fair question at matched size.

Results

Setup: same corpus, split, eval windows, steps, batch and context as C1; medians over seeds 0–2.

variant	parameters	bits/char
learned positions	1.82M	**2.309**
sinusoidal positions	1.80M	2.349
no positional encoding	1.80M	**2.641**
learned positions, at C1's budget	242k	2.508

Removing position costs **0.304 bits/char** (range 0.265–0.348) — the largest ablation in the part, and exactly what the equivariance argument predicts. Learned beats sinusoidal by 0.040 at this scale. The causal check returns **exactly 0.0** on every seed, for both the logit perturbation and the attention mass above the diagonal.

What surprised me — and the result I cannot explain away

Line up C1 and C3, which share every training setting:

model	parameters	bits/char
vanilla RNN (wide)	343k	**2.263**
transformer, learned PE	1.82M	2.309
LSTM	320k	2.334
transformer at C1's budget	242k	2.508

The wide vanilla RNN beats every transformer arm, using 5× fewer parameters than the best of them, and the parameter-matched transformer is the worst model in the comparison. That is not the story anyone expects in 2026, and it is not a claim that transformers are worse — it is a claim about this regime: 500 steps, 64 characters of context, 250k characters of training text, and *no warmup* for the transformer, which B5 showed is the one thing Adam-family optimizers are picky about at the start of training. Transformers are built to exploit long context and large data, and this rung deliberately gives them neither. The honest summary is that I measured a regime where the inductive bias of recurrence is worth more than the flexibility of attention, and that regime is small.



Figure 11: Positional ablation

Open threads

Warmup and longer context, the two runs this section called obvious, have been done in Part C at scale. Warmup changed nothing (2.310 → 2.312), and longer context did *not* make the recurrent models lose at a 49M-token budget. The transformer is the one model using the long context, but it was too under-trained to profit. Still open: RoPE instead of additive encodings. And KV caching, which is the difference between a sampler that is $O(T)$ per token and one that is $O(T^2)$.

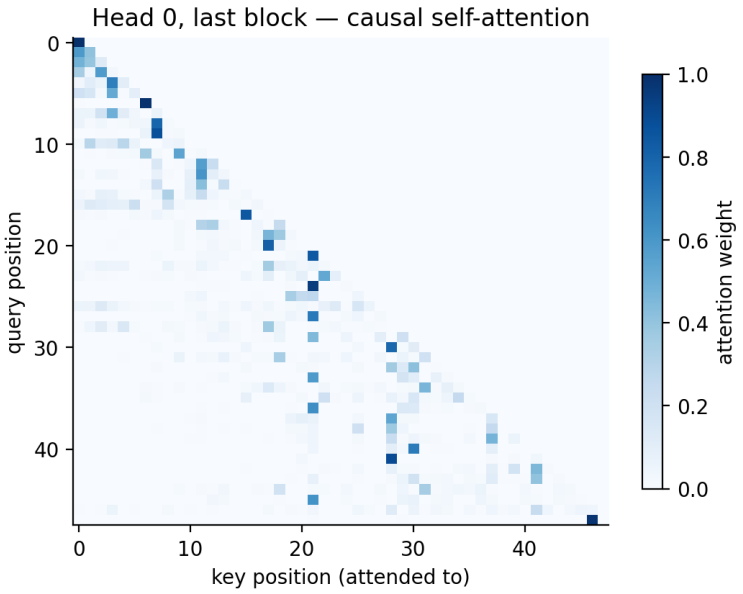


Figure 12: Attention map

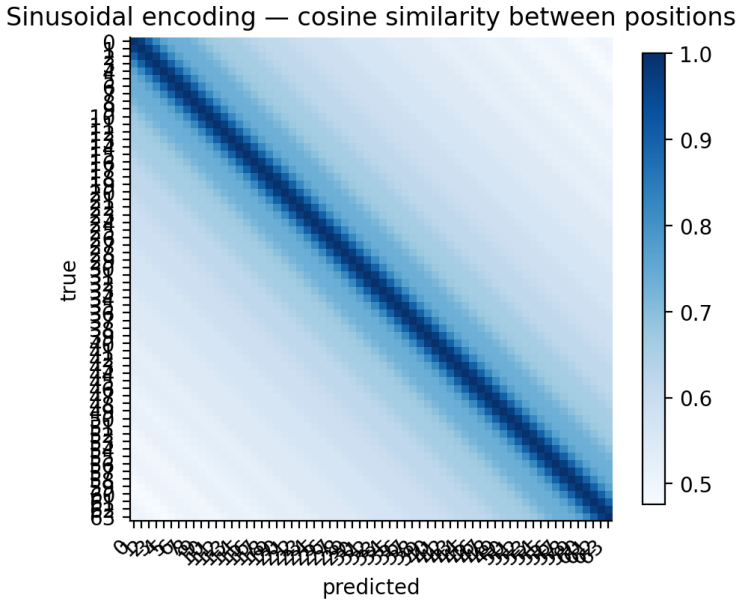


Figure 13: Positional similarity

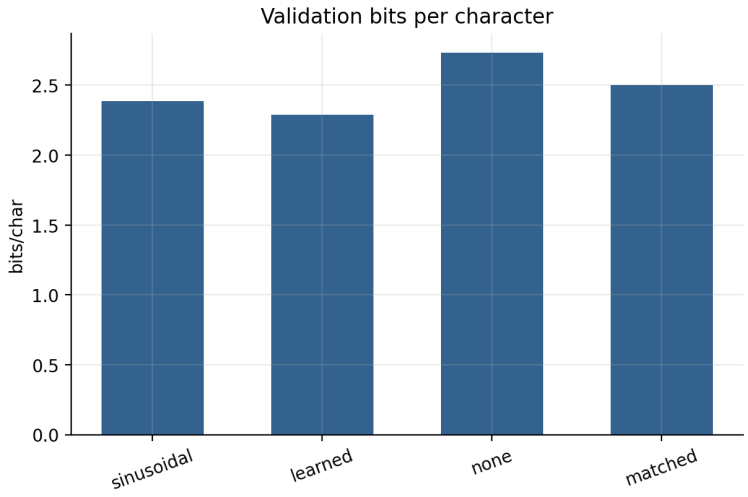


Figure 14: Bits per character

Further reading

Attention Is All You Need; The Annotated Transformer; Xiong et al., *On Layer Normalization in the Transformer Architecture* (pre-LN vs post-LN).

C4 — Fine-tuning a small ViT

Question. A ViT is C3's block applied to image patches. Without a CNN's built-in priors, how much does it depend on pretraining?

The idea

A **Vision Transformer (ViT)** is C3's transformer pointed at images. An image is cut into a grid of small patches (16×16 pixels in the pretrained model), each patch is flattened and linearly projected into a token, and the sequence of patch tokens goes through a standard transformer encoder. A special extra **[CLS] token** gathers information from all patches, and a linear layer on its output gives the class.

Unlike a CNN (B3), a ViT has no built-in assumption that nearby pixels belong together or that a cat in the corner is still a cat. It has to *learn* those facts from data, which makes it data-hungry. The standard remedy is **pretraining** on a huge dataset (ImageNet here), then reusing the model:

- **linear probe** — freeze the pretrained network and train only a new final layer. This asks whether the pretrained features already separate the new classes.
- **fine-tuning** — keep training the whole network on the new task, gently, with a small learning rate.

The image becomes a 16-token sentence plus a [CLS] *token. From there on the model is identical to a text transformer. The classification is read off the* [CLS] *token's final state.*

The math

Patchification is the only image-specific step. An image $x \in \mathbb{R}^{H \times W \times C}$ is cut into $N = HW/P^2$ patches of side P, each flattened and projected:

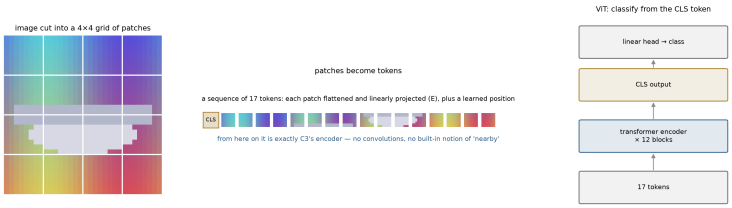


Figure 15: From an image to a sequence of tokens

$$z_0 = \left[x_{\text{cls}};\, x_p^1 E;\, x_p^2 E;\, \ldots;\, x_p^N E\right] + E_{\text{pos}}, \qquad E \in \mathbb{R}^{P^2 C \times d}.$$

After that it is C3's encoder verbatim. At 224^2 with $P = 16$ that is $N = 196$ tokens, and attention's $O(N^2)$ cost is why input resolution is expensive here in a way it is not for a CNN.

What the architecture does not assume. A convolution hard-codes locality (a unit sees a small neighbourhood) and translation equivariance ($f(\text{shift}(x)) = \text{shift}(f(x))$). Self-attention assumes neither: every patch can attend to every other from layer 1, and position enters only through E_{pos}, which is *learned*. That is more expressive and strictly less constrained — so with little data the ViT must learn from examples what a CNN gets for free, which is the standard explanation for ViTs being data-hungry and the thing this rung measures.

Probe versus fine-tune ask different questions. A linear probe on frozen features asks *are the pretrained representations already linearly separable for this task?* — it optimizes $\min_W \sum_i \ell(W\phi(x_i), y_i)$ with ϕ fixed, a convex problem in W. Fine-tuning asks *can the representation itself be improved?*, at 7000× the trainable parameters and the risk of destroying pretrained features with too large a step (hence the much smaller learning rate).

What we measure, and why

- **Test accuracy** on the four held-out classes (chance 0.25), for three arms: from scratch, linear probe, full fine-tune. The differences between arms are the point. *Probe − scratch* measures what pretraining contributed. *Fine-tune − probe* measures what adapting the features adds on top.
- **Trainable parameters** per arm — the probe trains 772 numbers and the fine-tune 5.5 million. That ratio is what makes the probe result striking.
- **Wall-clock seconds** on the T4 — the practical cost of each arm.
- **Confusion matrix** — which classes get mixed up (cars and trucks, as it turns out), which accuracy alone hides.

Results

Setup: ViT-Tiny, 4-class CIFAR subset — plane/car/ship/truck — 1500 train images, 2 epochs, medians over seeds 0–2.

arm	test accuracy	trainable parameters	seconds (T4)
from scratch	0.420	5.53M	12.8
linear probe	0.866	**772**	5.5
full fine-tune	**0.955**	5.53M	12.8

Pretraining is worth **+0.452** (probe over scratch, range 0.421–0.458); fine-tuning adds a further

+0.090 over the probe (range 0.073–0.100).

What surprised me

Not the ordering — the *ratio*. The linear probe trains **772 parameters**, four classes' worth of weights on a 192-dimensional feature, and lands within 9 points of a full fine-tune of 5.5 million. Essentially all of the task-relevant work was already done by ImageNet pretraining; the fine-tune is polishing. And the from-scratch arm at 0.420 is barely above the 0.25 chance of four classes — 1500 images is simply not enough to teach a transformer what an edge is.

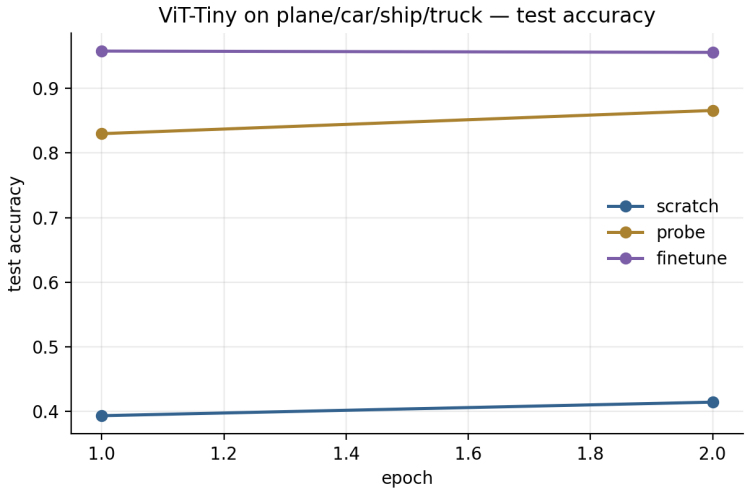


Figure 16: Arms

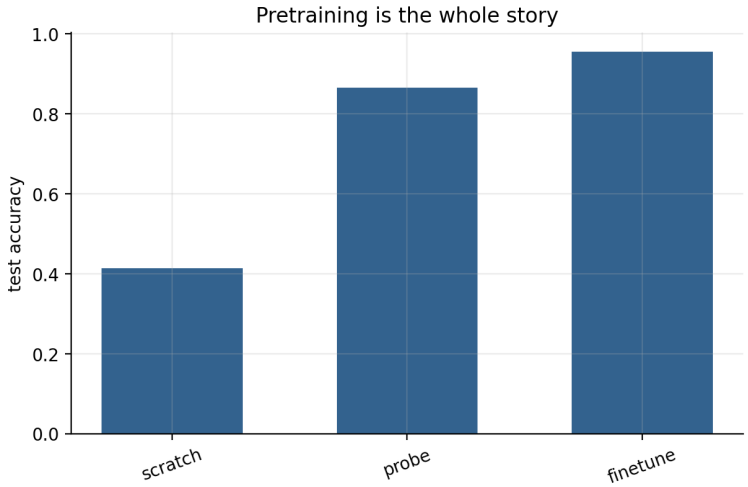


Figure 17: Final accuracy

Open threads

Freezing progressively more blocks, to find where "probe" turns into "fine-tune". Data augmentation, which is how the ViT paper's own from-scratch results were rescued. DINO or MAE features instead of supervised ImageNet ones — that comparison is part D's business.

Further reading

An Image is Worth 16x16 Words (ViT); Steiner et al., *How to train your ViT*; timm.

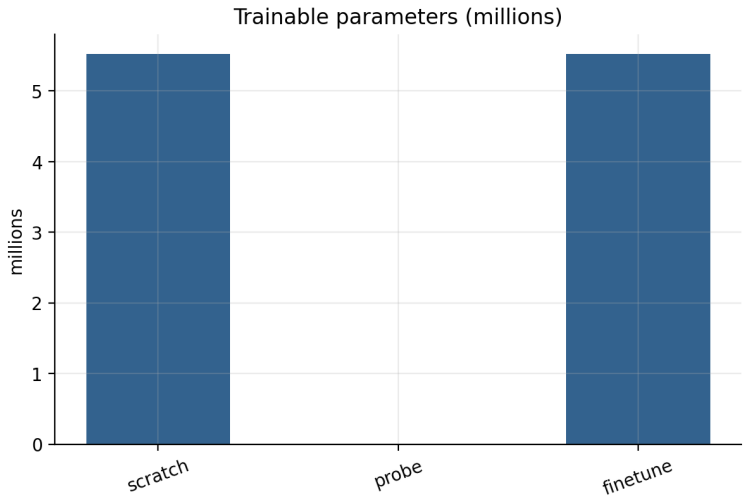


Figure 18: Trainable parameters

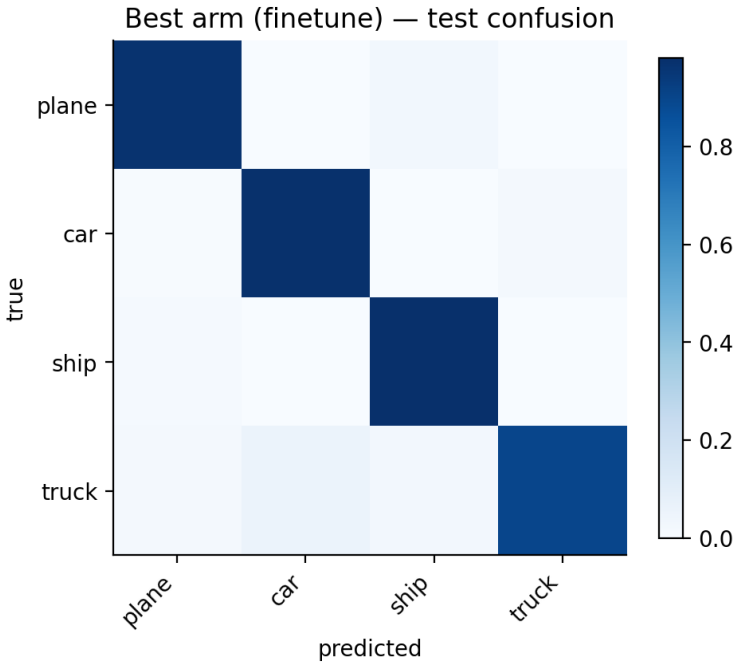


Figure 19: Confusion

C5 — What changed since C3, and what it cost

Question. C3's transformer is eight years old as a design. Every production decoder since has changed the attention layer. What are those changes actually buying, in a quantity that can be computed rather than argued about?

The idea

When a trained language model **generates** text, it produces one token at a time, and each new token attends over everything before it. Recomputing every earlier token's keys and values at every step would be wasteful, so they are stored: the **KV cache**. It grows by one entry per token per layer, and every new token has to read all of it. At long context, *memory*, not arithmetic, becomes the limit: how many bytes the cache takes, and how fast they can be read.

Every major decoder design since C3 has changed attention to shrink one factor of that cache:

- **Grouped-query attention (GQA)** — many query heads share a few key/value heads. Fewer K/V heads, proportionally smaller cache.
- **Sliding-window attention** — each layer only attends to the last W tokens, so its cache stops growing at W.
- **Multi-head latent attention (MLA)** — cache one compressed low-rank vector per token instead of full per-head keys and values, and expand it on the fly.
- **Linear attention / delta rule** — replace the cache with a fixed-size state matrix, updated at each token. This is an RNN again (C1), with a much larger state and a smarter update rule.

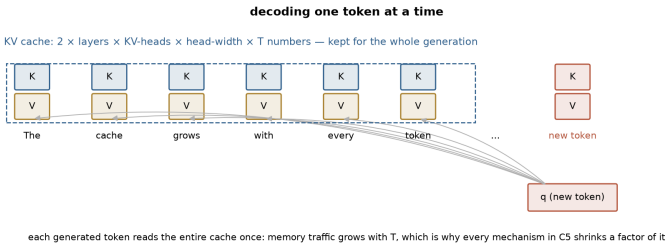


Figure 20: Decoding one token at a time

The new token's query must read every cached key and value. The cache's size is layers × KV heads × head width × context length, and each mechanism above removes or caps one of those factors.

This rung trains nothing. It reads the published configurations of real models and computes, exactly, what each design stores.

The math

The bill attention pays is not the quadratic term. At generation time a decoder emits one token at a time, and each new token attends over everything already produced — so every past key and value must be *retained* and re-read. The cache for a multi-head stack is

$$M_{\mathrm{KV}} = 2\, b\, L\, H\, d_h\, T,$$

bytes for L layers, H heads of width d_h, context T and b bytes per element. It grows linearly with context and is read in full per token, which makes decode memory-bound: the arithmetic is trivial next to the bandwidth. Four families of change attack one factor each.

Grouped-query attention shrinks H to H_{kv} in that formula and nowhere else — queries keep their full head count, so the model's capacity to *ask* different questions is untouched while the memory of answers is shared.

Sliding-window attention replaces T with $\min(T, W)$, making a layer's cache constant past the window. Reach is not lost, it is relocated: stacking L such layers gives a receptive field of $L \cdot W$ through composition, the same trick a CNN uses for its effective field.

Multi-head latent attention stops caching per-head keys and values at all. DeepSeek-V2 projects them into one shared low-rank latent c_{KV} of rank r, plus a small decoupled rotary key that must stay outside the compression because RoPE does not commute with the down-projection:

$$M_{\text{MLA}} = b\, L\, (r + d_{\text{rope}})\, T.$$

The head count has left the expression entirely.

A delta rule leaves the cache paradigm. Gated DeltaNet keeps a fixed-size matrix $S \in \mathbb{R}^{H_v \times d_k \times d_v}$ updated by an outer-product rule, plus a short causal convolution window. Its size is chosen by the architect, not by how much text has been read — so the state is **constant in** T. That is C1's bargain exactly: compress the past into a fixed state. What changed is not the bargain but the budget, and the update rule that fills it.

What we measure, and why

- **Decode state in MiB at a given context length** (1k to 256k tokens, fp16) — everything the model must keep in memory to generate the next token. It is the quantity that decides whether a long context fits on a GPU at all, and at long context it can exceed the model's own weights.
- **Shrink factor per mechanism**, each measured against *its own model with that one mechanism removed*. Comparing two different models would credit a mechanism with differences that come from ten other design choices.
- **Crossover context** — for the fixed-state design, the context length at which its constant state becomes cheaper than a growing cache.
- **Parameter counts checked against published figures** — validation that the accounting code reads configs correctly before its other numbers are trusted.

What I built

No training. Five configurations — C3's own, measured from this repo, and four published ones recorded with their hub ids — reduced to layer lists and evaluated for decode state against context. Each mechanism is also ablated **against its own config with that mechanism removed**, rather than against a different model, so the number attributed to it is not really a difference in ten other things. The accounting lives in `src/mlladder/archspec.py` and is shared with H6.

Results

Setup: decode state in MiB, fp16.

decoder	layers	1k	16k	64k	256k
C3 (this repo)	4 full	3.0	48	192	768
Qwen3-8B	36 full (GQA 32:8)	144	2,304	9,216	36,864
Mistral-7B	32 sliding (W=4096)	128	**512**	**512**	**512**
DeepSeek-V2-Lite	27 MLA (r=512)	30	486	1,944	7,776
Qwen3-Next-80B-A3B	36 linear + 12 full	62	422	1,574	6,182

Each mechanism against its own config without it, at 64k tokens:

mechanism	cache shrink
GQA (32 query heads over 8 KV)	**4.0×**
sliding window 4096	**16.0×**
MLA (rank 512 + 64 rope)	**10.7×**
delta rule (vs its own all-attention twin)	**3.9×**

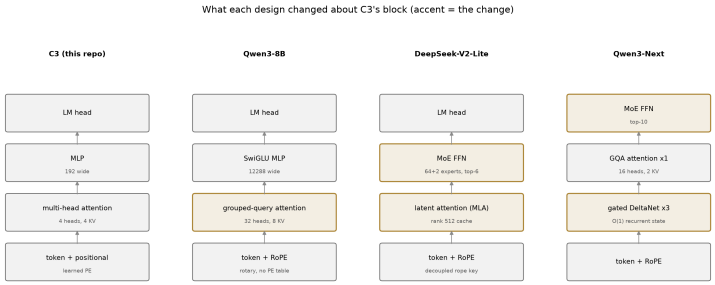


Figure 21: Architectures

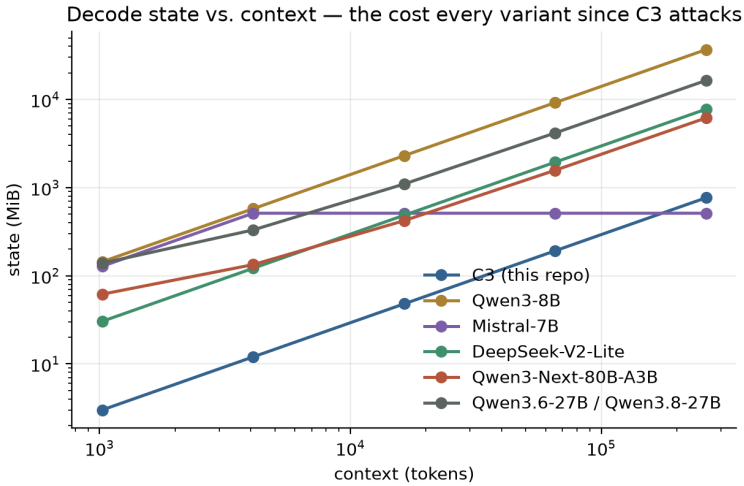


Figure 22: Decode state

What surprised me

Three things.

The numbers are enormous and they are not about parameters. Qwen3-8B carries **9.2 GiB of KV cache at 64k tokens** — larger than the 8B model's own weights in fp16. The thing that stops a decoder

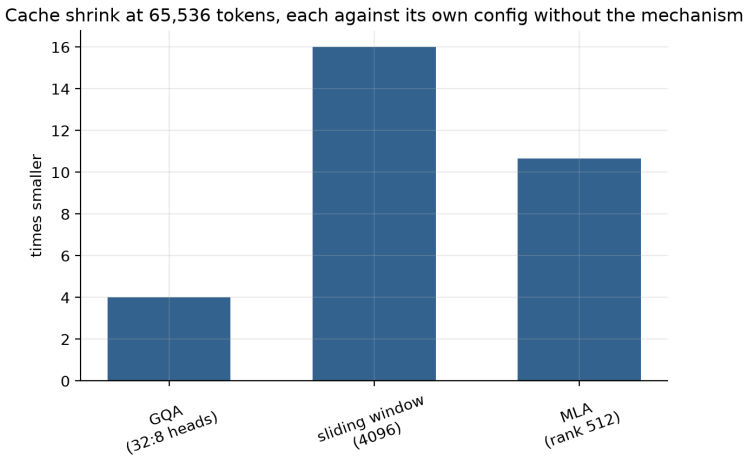


Figure 23: Mechanism savings

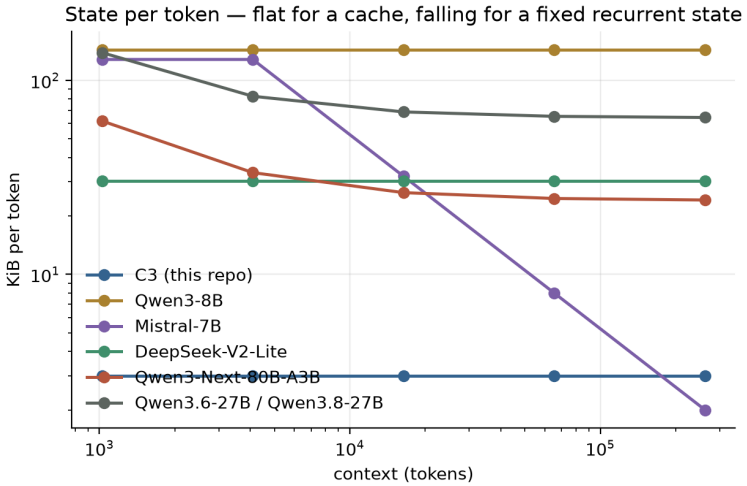


Figure 24: State per token

serving long contexts is usually not its size; it is the cache the context creates, and no amount of quantizing weights touches it.

Mistral's window is the bluntest instrument and the biggest win. A cache that simply stops growing beats every clever compression asymptotically: 512 MiB at 64k and still 512 MiB at 256k, while MLA's 10.7× compression is still growing linearly underneath it. And the reach is not obviously sacrificed — 32 layers at a 4096 window compose to a 131,072-token receptive field. The cost is that the reach is *indirect*, mediated through depth, which is exactly the compromise C1's recurrence made and lost on. Whether it loses here is not something state accounting can answer.

C1's answer came back, and the crossover is early. The delta-rule state is 1.05 MiB per linear layer, so Qwen3-Next carries **38 MiB before reading a single token** — strictly worse than any cache at short context. It pays that back against its own all-attention twin from **537 tokens** onward. I had expected a constant-state design to need a very long context to justify itself; at these dimensions it needs about two paragraphs. The tempting comparison — the hybrid against Qwen3-8B, which crosses at 322 tokens — is the one to avoid, because it credits the delta rule with savings that are really Qwen3-Next's much more aggressive GQA (2 KV heads against 8).

A check worth having. DeepSeek publishes V2-Lite at 15.7B total and 2.4B activated. The accounting here computes 15.8B (**0.6% off**) and, excluding the embedding and output head that DeepSeek's transformer-only figure omits, 2.32B (**3.2% off**). H6's independent check against

TrOCR agreed to 1.3%. Two validations on models with known answers is what makes the other numbers in these tables worth reading.

Open threads

Gemma 4's ring-buffered sliding KV, which sizes the buffer at window + prefill chunk rather than the full context and is implemented in the accompanying fork — the natural next arm. Decode throughput *has* been measured for four real decoders (below), but not against these predictions: that needs cache bytes divided by the card's bandwidth, compared with tokens/s at several context lengths, and the runs below used one short prompt each. And quality: none of this says whether a 512 MiB windowed cache answers questions as well as a 9 GiB full one, which is the question that decides anything.

Further reading

GQA; Mistral 7B for sliding windows; DeepSeek-V2 for MLA; RoFormer; and candle's `qwen3_5.rs`, `deepseek2.rs` and `gemma4/` for the implementations.

Part C at scale: more context, more data, a bigger transformer

C1 and C3 compared recurrence and attention at 64 characters of context, 250k characters of text and 500 steps, and the wide vanilla RNN won. C3 said why that was not the end of the story and listed the runs that would settle it: warmup for the transformer, and longer context "where the recurrent models must lose". C1 listed GRUs. Those runs are here, on a Kaggle T4, driven by `tools/gpu/c_context_scaling.py`. Raw values are in `data/c-gpu.json`. Every number is a median over seeds 0–2.

The idea

Three questions, three experiments:

1. **Was C3's loss just the missing warmup?** Rerun C1/C3's exact regime (same corpus slice, context, batch, steps, peak learning rate, the hand-written C1 cells and the C3 transformer), with and without linear warmup plus cosine decay.
2. **Does attention win once there is context worth attending to?** Train on **enwik8**, the standard byte-level benchmark: the first 100 MB of an English Wikipedia dump, markup included, with long-range structure (tags that open and close hundreds of bytes apart, repeated names, tables). Five models: a vanilla RNN, a GRU and an LSTM, each ~4.5M parameters, a transformer matched to them (~4M), and a **bigger transformer (~25M)**. Each is trained at 64, 256 and 1024 bytes of context, on the **same 16,384 tokens per step** (so a longer context means fewer sequences per batch, not more data), for 3000 steps: 49M tokens, about half an epoch.
3. **Can each architecture remember at all?** A **copy task**: n random symbols, then the same n again. The second half is predictable only by looking exactly n positions back, so accuracy there measures usable memory and nothing else.

The recurrences use cuDNN's `nn.RNN/GRU/LSTM` (the same equations as C1's cells, which would take hours as a Python loop at 1024 steps). The transformer is C3's own class with attention routed through PyTorch's fused kernel, checked against C3's explicit softmax to 1.2×10^{-7} before any training.

What we measure, and why

- **Bits per byte** on 256 KiB of held-out enwik8: cross-entropy in base 2, the enwik8 convention (C1's bits per character, for bytes). For scale: Graves' 2013 LSTM reached 1.67, later recurrent models about 1.25, and large transformers below 1.0.
- **Loss by position in the window** — the direct measure of *using* context. The prediction at position t has t bytes of history, so a model that exploits long context keeps improving as t grows, and one that does not goes flat.
- **Copy-task accuracy on the repeated half** (chance $1/32$).
- **Training throughput** (tokens per second, fp16, T4) — what each design costs as context grows.

Results

1. Warmup does not rescue C3's transformer.

model	constant LR (C1/C3's setting)	warmup + cosine
vanilla RNN, wide	2.269	**2.251**
LSTM	2.319	2.448
transformer	2.310	2.312

C3's leading explanation for its result was wrong: with warmup the transformer scores 2.312 against 2.310 without it. The wide RNN still wins, and cosine decay *hurts* the LSTM by 0.13 bits. At 500 steps, a schedule that spends its second half at a lower learning rate costs the slowest learner the most.

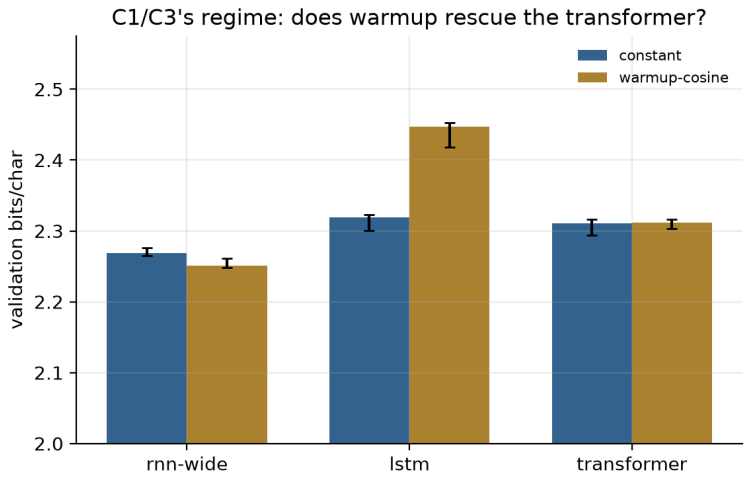


Figure 25: Warmup in C1/C3's regime

2. Memory: the transformer copies 512 symbols back perfectly; the LSTM cannot copy 128.

n	LSTM (4.5M)	transformer (4M)
32	0.903	**1.000**
128	0.031 (chance)	**1.000**
512	0.031 (chance)	**0.999**

This is the cleanest result in the section. A recurrent model has to squeeze n random symbols (5 bits each) through its fixed state and fails once that state is too small. Attention simply looks back n positions, whatever n is. C2's bottleneck, measured on the architecture that removed it.

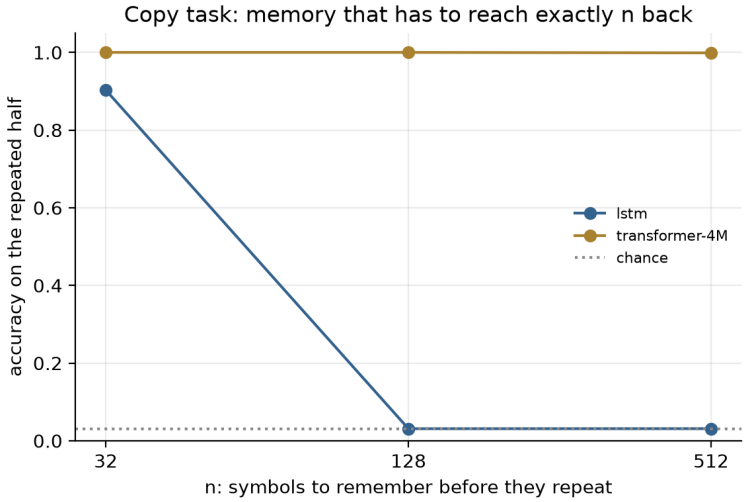


Figure 26: Copy task

3. enwik8: the bigger transformer is the best model, but more context did not help it at this budget.

context	RNN	GRU	LSTM	transformer 4M	**transformer 25M**
64	1.854	1.795	1.883	1.837	**1.718**
256	1.791	1.701	1.838	1.863	**1.668**
1024	1.804	**1.687**	1.848	2.043	1.721

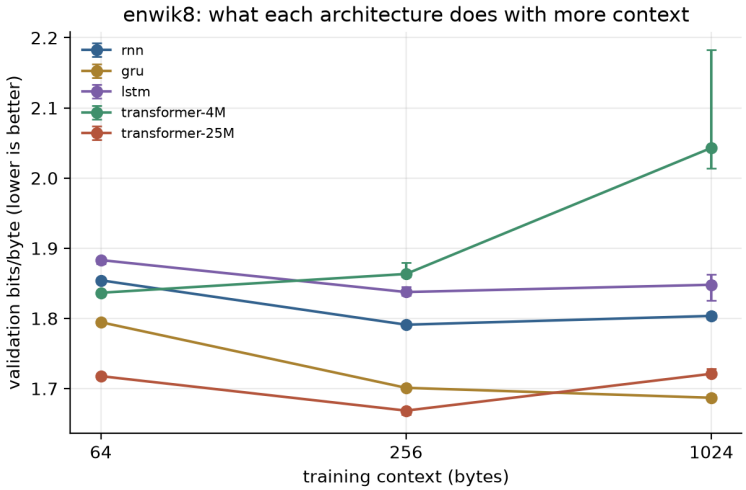


Figure 27: Bits per byte against context

The 25M transformer posts the best number in the table, 1.668 at 256 bytes. Among the size-matched ~4M models, however, the **GRU wins at every context**, and the matched transformer gets *worse* as context grows, ending unstable at 1024 (seed range 2.01–2.18). So the promised crossover, "longer context, where the recurrent models must lose", did not happen at this budget, and the bigger transformer lost ground from 256 to 1024.

The loss-by-position curve says why, and it is not "attention cannot use the context":

trained at 1024	positions 64–256	256–512	768–1024
GRU	1.675	1.687	1.666
transformer 25M	1.751	1.722	**1.658**
transformer 25M trained at 256, same positions	*1.622*	—	—



Figure 28: Loss by position

The GRU is **flat** from byte 64 to byte 1024: whatever it keeps in its state, the extra 900 bytes add nothing. The transformer is the only model still improving deep into the window (−0.093 bits from the early positions to the late ones), so it *is* reading the long context. But the model trained at 1024 is worse than its 256-trained twin *at the same short positions* (1.751 against 1.622). At a fixed 16k tokens per step, a 1024-byte context means 16 sequences per batch instead of 64, and 1024 learned positions to fit instead of 256, all from 49M tokens, which is **two tokens per parameter** for the 25M model against Chinchilla's ~20 (G1). The transformer is under-trained, and the longer context makes that worse before it helps.

Throughput. At 64 bytes the recurrences are fastest (RNN 504k tokens/s against the matched transformer's 348k). At 1024 the order flips: the matched transformer trains at **254k tokens/s** against 182–202k for all three recurrences, even with cuDNN, because a recurrence must walk its sequence one step at a time while attention's quadratic cost is still cheap at this length. The 25M transformer runs at 67–83k tokens/s throughout.

What surprised me

C3 closed on a prediction ("longer context, where the recurrent models must lose"), and at a realistic-looking budget it failed for the matched models. The GRU beat a same-size transformer at 1024 bytes of context, on the benchmark where transformers made their name, and the transformer's own curve went the wrong way. Two things turned that from a contradiction into a diagnosis: the per-position loss, which shows the transformer is the one model that *uses* the long context, and the copy task, which shows its memory is not the problem. What is left is budget. The 4× longer run below tests that directly.

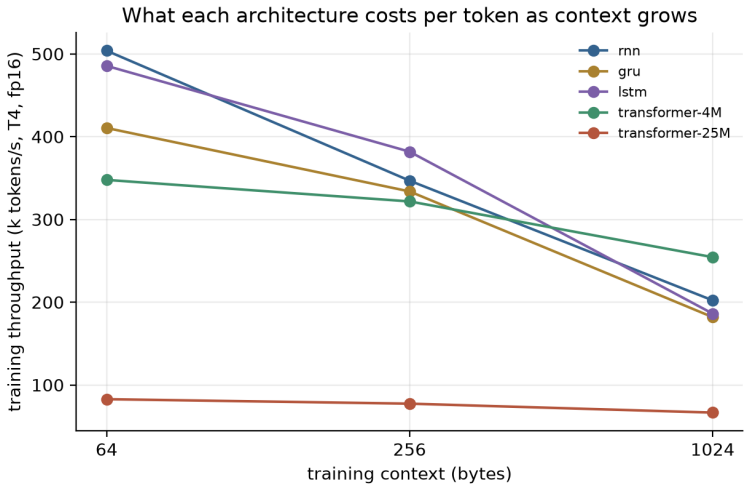


Figure 29: Throughput against context

Open threads

A tokenised corpus, since byte-level modelling spends most of a small model's capacity on spelling. RoPE instead of learned positions, which do not have to be learned separately at every offset. And a learning-rate sweep for the matched transformer at 1024, whose seed spread says the setting is on an edge.

The decoders, actually run

C5's method is reading published configs and computing decode state from them. That method has two failure modes it cannot detect from the inside: the arithmetic can be wrong, and the *config* can fail to contain the thing you are asking about. Loading the weights finds both.

Everything below ran on one Colab T4 (15.4 GB), which is the binding constraint and the reason quantization is not an optional extra here. Raw values in `data/c-real-models.json`; the scripts are in `tools/real_models/`.

model	how it fit	resident	gen tok/s
DeepSeek-V2-Lite (15.7B)	bitsandbytes NF4, double-quantised scales	9.62 GB	4.44
Gemma-4-E2B-it	bitsandbytes NF4	6.70 GB	6.35
Qwen3.6-27B	GGUF UD-IQ3_XXS, 3.06 bpw, fully offloaded	12.0 GB	10.2
Qwen3.8-27B	GGUF UD-IQ3_XXS, 3.06 bpw, fully offloaded	10.9 GB	9.5

All four produce coherent text. Two things came out of it that the arithmetic could not have.

A parameter count read off a quantized model is not a parameter count. `sum(p.numel())` reports **8.06B** for DeepSeek-V2-Lite, a published 15.7B-parameter model, because bitsandbytes

packs two 4-bit weights into each byte. C5's own count of 15.8B is right and the loaded model's `numel` is wrong by exactly 2x — which is the direction that would have been very easy to believe, since "the config over-counts" is a more plausible-sounding story than "the tensor told me half."

Gemma-4-E2B-it runs one KV head for eight query heads. That is multi-query attention — the limit of the GQA lever, not a step along it. C5 prices GQA at Qwen3-8B's 32:8, a 4x cache reduction; MQA at 8:1 is 8x, and the reason the family stops there is that there is nothing left to share.

Two generations, one architecture

The interesting result is a negative one. Asked to compare Qwen3.6 against Qwen3.8, the size-matched pair is `Qwen3.6-27B` and `Qwen3.8-27B`. Their configs were flattened and compared key by key:

61 keys compared. One differs. It is `transformers_version` — the library that wrote the file.

Same 64 layers, same 5120 hidden, same 24 query heads over 4 KV heads, same 256-wide head, same 17408 intermediate, same 248320 vocabulary, same interleave of 48 linear-attention layers with 16 full-attention ones at interval 4, same single MTP layer. Both ship **55.56 GB** of bf16 weights — identical to the hundredth of a gigabyte, so the parameter counts match too.

Whatever separates these two generations at 27B, it is weights, data or post-training. It is not shape, and **no amount of reading configs will find it** — which is exactly the boundary of what this rung can see, now measured rather than assumed. C5 carries them as a single entry for that reason, with a test that fails if a later edit splits them without an actual difference to point at.

The runtime behaviour is consistent with that: 10.2 against 9.5 generated tokens/s at the same quantization. I am *not* reporting that 0.7 as a difference between the models, because the two GGUF files are 12.0 GB and 10.9 GB at the same nominal 3.06 bpw — the quantizer allocated bits differently between them, so the arms are not matched. A throughput comparison across two different quantizations of two models is a comparison of the quantizations.

One qualitative note, offered as an anecdote and not a result: at a 96-token budget **neither model produced an answer**. Both are reasoning models, both opened a thinking block, and both were still in it when the budget ran out — 3.6 with a numbered plan, 3.8 telegraphically. n=1 at temperature 0 on one prompt is worth exactly what this paragraph claims for it, which is nothing except that the budget was wrong for the model class.

What did not run. `Qwen3.8-Flash-Next` is 360 GB of weights. The 27B pair at 4-bit AWQ is 20.4 and 21.0 GB against a 15.4 GB card, which is why the 3-bit GGUF route is what made them runnable at all — the quantization was not a convenience, it was the difference between a measurement and a paragraph explaining why there wasn't one.

Through the candle fork

Part H's OCR model runs through the accompanying candle fork and matches transformers to sixteen digits. The decoders went less well, and the failures are specific enough to be useful.

Gemma-4-E2B-it needed two patches before it would load at all. The fork's `TextModel` reads the decoder from `model.*`; a multimodal checkpoint keeps the same decoder at `model.language_model.*` beside its vision and audio towers, so text-only loading died at `model.embed_tokens.weight`. And the example hardcodes bf16 on GPU, which needs sm_80.

With a nested-prefix constructor and a `--dtype` flag it loads and generates at **25.7 tok/s** — and emits the word "thought" repeated to the token limit.

That is worth one control before blaming anyone. Transformers at **f16**, same model, same card, same prompt and chat template, answers correctly at 9.63 tok/s — so it is not f16 precision. And the prefix patch is behaviour-preserving: the constructor's other uses of the unprefixed builder are `dtype()` and `device()`, and the one root-level weight (`lm_head`) is never reached because this config sets `tie_word_embeddings: true`.

I concluded from that that the fork's `gemma4` f16 path was wrong on sm_75. **That conclusion was wrong, and the answer was already written down in the repository I had not read yet.** The catalog's `candle-llm-safetensors` executor carries this comment:

> Some checkpoints ship a tokenizer.json without a BOS post-processor (e.g. gemma-4-e2b-it) — gemma-family models degenerate into token copy-loops without it.

A copy-loop is exactly what I saw, on exactly that checkpoint. I had reached for "the kernels are wrong on this card" when the available explanation was a missing token — and the thing that distinguishes them is that one of them was already diagnosed by someone who had run the model more than twice.

With the same BOS prepend added to the example, on the same binary, the same card, the same f16 and the same chat template:

> Grouped-query attention reduces KV cache size by grouping queries, which allows the model to share key and value states across multiple queries. This sharing significantly reduces the total number of unique key and value vectors that need to be stored in the cache for a given sequence length.

5.89 tok/s with the template, 24.55 on a bare prompt. **"thought" repeated to the token limit before the prepend, a correct answer after** — a controlled pair where the only difference is one token, which is what it takes to tell a missing BOS from a broken kernel. The f16 accusation is withdrawn.

Two of the three patches Gemma-4 needed, incidentally, are things `qwen3_5.rs` already does for itself in the same fork: it detects the `model.language_model` prefix rather than assuming `model.`, and it already has load-time quantization. `gemma4` is the sibling that fell behind.

DeepSeek-V2-Lite needed the quantization the example did not have. It is now wired: the same load-time `Proj` that `gemma4` and `qwen3_5` use — quantize on the CPU so a full f32 copy never reaches the GPU, keep a `QMatMul` — threaded through attention, the MLP, the 64 MoE experts and `lm_head`, behind `DeepSeekV2::new_with_quant` and a `--quant` flag. One detail the model forces: k-quants need the reduction dimension to be a multiple of 256, and DeepSeek's dense `down_proj` reduces over 10944 while each expert's reduces over 1408. Neither qualifies, both are multiples of 32, so those layers fall back to Q8_0 rather than silently producing a broken tensor.

It compiles and the flags are live on the binary, and that is all I can say for it — the GPU session was reclaimed during the 31.4 GB download, twice, so no token has been generated through candle's quantizer. Wiring that has never run is a claim, not a result, and it is recorded here as one.

What *did* run is the same model pre-quantized on the hub (`slowfastai/DeepSeek-V2-Lite-bnb-4bit`), and the comparison is the useful part:

	load	resident	gen tok/s
quantize the f16 weights at load	189.8s	9.62 GB	4.44
download it already quantized	**119.1s**	9.62 GB	4.49

Same footprint, same throughput, 70 seconds sooner. **Pre-quantizing buys load time and nothing else** — which is worth knowing before treating an ahead-of-time quantized checkpoint as a performance decision rather than a convenience one.

Qwen3.6/3.8 had `qwen3_5.rs` in `candle-transformers` and no binary to invoke it, so that binary is now written: `--quant q8_0..q2k`, `--dtype`, and Qwen's chat template. **`Qwen/Qwen3.5-4B` at q4k answers correctly through it at 26.79 tok/s**, which verifies the example and the quantization path together.

Getting there needed one fact about the catalogue. Checking all 465 models under the Qwen org: **the Qwen3.6 generation ships no small checkpoint** — 27B and 35B-A3B, and their FP8 twins, and nothing else. Qwen3.5 ships 0.8B, 2B, 4B, 9B, 27B and up. So there is no official small Qwen3.6 to verify against, and the 27B's 55.6 GB will not fit a 50-minute session alongside a from-scratch build.

The nearest small thing is a community REAP-pruned derivative of 35B-A3B, `inference-optimization/Qwen3.6-8B-A1.6B`. It loads and generates — and emits multilingual token soup at 5.23 tok/s. Qwen3.5-4B through **the same binary, the same flags and the same quantization** answers correctly, so the failure belongs to the pruned derivative, not to the example. Which is the whole reason to run a control before concluding anything: without it, "my new example produces garbage" would have been the obvious reading, and it would have been wrong — the same mistake I made with Gemma's missing BOS, caught this time.

What part C settled

1. **Every architecture here is a different answer to "how does information travel?"** Recurrence carries it in a state and pays multiplicatively per step; gating buys an additive path; attention refuses to compress and pays $O(T^2)$. The measurements line up with the algebra: the RNN's geometric Jacobian product shows up as a 27-step horizon, the LSTM's product of forget gates as 63, and attention's convex combination as a bottleneck-vs-attention gap that grows from 0.003 to 0.98 as the source gets longer.
2. **Reach is not the same as use.** The LSTM's gradient arrives 63 steps back and it still cannot solve 30-step recall at this budget, and its extra memory *loses* to a wide vanilla RNN on the actual language-modelling task. An architecture's capability and what a training run extracts from it are different quantities.
3. **The famous architecture lost, and the regime is why.** A 343k-parameter vanilla RNN beat a 1.8M-parameter transformer on 250k characters at 64 tokens of context with no warmup. Nothing about that contradicts the literature; it locates the boundary of it, which is more useful than a fourth reproduction of a result everyone already believes.
4. **Two metrics on the same run told opposite stories.** C2's bottleneck model keeps 71% character accuracy while getting 0.5% of dates right. Choosing the metric that matches the task's actual failure mode did more work here than any modelling decision.

5. **A seed does not pin a number, and which numbers it fails to pin is informative.** Across three full sweeps — one on CPU, two on separate T4 sessions — C1, C3 and C4 reproduced their medians *exactly* (bits/char, horizons, accuracies, the causal-leak zero). C2's bottleneck exact-match did not: 0.363, 0.349 and 0.326 at 101 characters, 0.012, 0.017 and 0.005 at 193. Seeding does not make cuDNN's backward deterministic, and the bottleneck arm sits exactly at the edge of failing, where an all-or-nothing metric over ten characters amplifies any tiny difference in weights. Its character accuracy, measuring the same runs, moved by only ~0.015. So the metric that made the failure legible is also the least stable one — both facts follow from it being a threshold measure near a threshold. The conclusion (the bottleneck collapses while attention does not) is unaffected on every run; the third decimal place is fiction. Figures and metrics here therefore come from one sweep on one device, and `--check` should be run on that device.

6. **The design C1 lost with came back, and the ladder is why I noticed.** C5's delta-rule stack is C1's bargain — compress the past into a fixed state — with a larger budget and a better update rule, and it pays for itself from 537 tokens of context. Reading Qwen3-Next's config as "a linear-attention hybrid" says nothing; reading it as "C1, with the compression problem solved differently" is only available because this part built C1 first.

Cost of the part: ~2,300 lines including tests, 27 test cases, and a full three-seed sweep in about 8 minutes on a T4 (C1 67s, C2 25s, C3 33s, C4 35s per seed; C5 is arithmetic) — the first part where the CPU-only path stopped being comfortable.

Figures regenerate with `uv run python tools/writeup_figures.py C`; the numbers they are quoted against live in `data/c-metrics.json`. The part README is the short version of this document.